THE TOTAL SITE OF A SIMPLICIAL SITE

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-simplicial-site-site (source lines 796-811); the proof environment begins two lines below it and is its own proof, closing at source line 832.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the simplicial site and its structure morphisms are the author's surrounding development in the same chapter.

Editorial note. The proof closes with the author's own remark that one claim follows by the same reasoning with details omitted; the retained package reproduces the author's proof exactly and does not supply the omitted details.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-spaces-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Spaces chapter, counted once: a simplicial object in the category of sites determines a total site in the described way.

Allowed standard prerequisites (editorial).

- Site and its axioms. A category with a notion of covering of each object, stable under base change, containing isomorphisms, and closed under composition and refinement.. The proof checks the axioms, in particular stability under base change, for the described coverings (source lines 818-831).
- Simplicial object in the category of sites. A sequence of sites with face and degeneracy functors, whose structure morphisms transform coverings into coverings.. The structure morphisms are what allow the pullback of a covering along a morphism of the total site (source lines 814-829).
- The total site of a simplicial site. Objects are the objects of the individual levels, morphisms combine a simplex map with a morphism into the induced pullback, and coverings are levelwise coverings with identity simplex map.. It is the data whose site axioms are verified in the proof (source lines 798-810).

Lemma 0.1. *Let $\mathcal{C}$ be a simplicial object in the category of sites. With notation as above we construct a site $\mathcal{C}_{total}$ as follows.*

(1) *An object of $\mathcal{C}_{total}$ is an object U of $\mathcal{C}_n$ for some n,*

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(2) *a morphism* $(\varphi, f) : U \to V$ *of* $\mathcal{C}_{total}$ *is given by a map* $\varphi : [m] \to [n]$ *with* $U \in \mathrm{Ob}(\mathcal{C}_n)$, $V \in \mathrm{Ob}(\mathcal{C}_m)$ *and a morphism* $f : U \to u_\varphi(V)$ *of* $\mathcal{C}_n$, *and*
(3) *a covering* $\{(id, f_i) : U_i \to U\}$ *in* $\mathcal{C}_{total}$ *is given by an* n *and a covering* $\{f_i : U_i \to U\}$ *of* $\mathcal{C}_n$.

Proof. Composition of $(\varphi, f) : U \to V$ with $(\psi, g) : V \to W$ is given by $(\varphi \circ \psi, u_\varphi(g) \circ f)$. This uses that $u_\varphi \circ u_\psi = u_{\varphi\circ\psi}$.

Let $\{(\mathrm{id}, f_i) : U_i \to U\}$ be a covering as in (3) and let $(\varphi, g) : W \to U$ be a morphism with $W \in \mathrm{Ob}(\mathcal{C}_m)$. We claim that

$$W \times_{(\varphi,g),U,(\mathrm{id},f_i)} U_i = W \times_{g,u_\varphi(U),u_\varphi(f_i)} u_\varphi(U_i)$$

in the category $\mathcal{C}_{total}$. This makes sense as by our definition of morphisms of sites, the required fibre products in $\mathcal{C}_m$ exist since u_φ transforms coverings into coverings. The same reasoning implies the claim (details omitted). Thus we see that the collection of coverings is stable under base change. The other axioms of a site are immediate. □